

Zhang Yichen

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EDUCATION

Tsinghua University

Ph.D. Candidate in Chemical Engineering, GPA: 3.99/4.00, Ranking: 3/86

June 2023 - Present

Xiamen University

B.Sc. in Science in Chemical Engineering, GPA: 3.82/4.00, Ranking: 1/42

Sep 2019 - June 2023

SCHOLARSHIPS & AWARDS

Friends of Tsinghua – Shanghai Huayi Scholarship, THU	Oct 2025
Best Poster Award, 22 nd National Electrochemical Congress	Oct 2024
Teng-Teng Scholarship, THU	Oct 2024
Gao-Yingshi Scholarship, THU	Oct 2024
Future Scholars Scholarship, THU	Sep 2023
National scholarship, XMU	Nov 2022
National scholarship, XMU	Nov 2021
Maotai scholarship, XMU	Nov 2020

SKILLS AND INTERESTS

Interests	Experimental and Theoretical Studies on Non-precious Metal Catalysts for ORR Structure-Activity Relationships of Catalysts with Atomically Dispersed Sites in Surface Catalytic Processes
Software	VASP, Material Studio, Gaussian, CP2K, Demeter, FDMNES, Jade, Advantage, CAD, SolidWorks

RESEARCH PUBLICATION

‡ = co-first authorship; * = corresponding.

1. **Zhang, Y.‡**; Gao, D.‡; Ouyang, D.; Yan, B.; Zhao, X.* Efficient harvesting of electricity, aromatic aldehydes and H₂ from lignin over nanoflower-like cobalt-based bifunctional electrocatalysts. *Green Chem.*, **2025**, 27, 8517-8531.

Overview: This work demonstrates a self-powered electrocatalytic system using nanoflower-like Co-catalysts for integrated lignin valorization. Via high-current electrodeposition on Ni foam, the bifunctional catalyst simultaneously harvests electricity (196.1 mW cm⁻² peak power density), aromatic aldehydes (2.90-3.34% yield), and H₂ (12.7 mL cm⁻²) from lignin. The direct lignin fuel cell drives electrolytic depolymerization and water splitting without external power, enabling sustainable production of chemicals and green hydrogen through efficient electron transfer from biomass.

2. Qiu, Y.‡; **Zhang, Y.‡**; Zhang, R.; Huang, M.; Tan, K. B.; Zhan, G.; Li, Q.; Huang, J.; Fu, G.* Internally hollow Cu₂O nanoframes with the abundance of {110} facets enhance direct propylene epoxidation. *Nat. Commun.*, **2025**, 16, 7802.

Overview: Engineered Cu₂O nanoframes via ligand-protected facet-etching achieve 0.18 mmol g_{cat}⁻¹ h⁻¹ PO formation rate with 83.8% selectivity at 175°C. The catalyst maintains performance >300 min, with mechanistic studies (DFT/characterization) confirming Cu₂O {110} facets promote active oxygen species generation and selective epoxidation.

3. Ran, L.‡; **Zhang, Y.‡**; Tong, W.; Chen, L.; Wang, M.; Zhou, H.; Farràs, P.*; Chen, S.*; Qiu, X.* Circumventing Radical Generation on Fe-V Atomic Pair Catalyst for Robust Oxygen Reduction and Zinc-Air Batteries. *Angew. Chem. Int. Ed.*, **2025**, 64, e202514542.

Overview: Established an ORR pathway-switching strategy using DFT-guided Fe₁V₁-NC atomic pair catalysts to circumvent ROS generation. The N₂-Fe-N₂-V-N₂ configuration enables direct O=O bond cleavage (side-on adsorption), achieving record stability (16 mV decay @ 50k cycles) and activity ($E_{1/2} = 0.89$ V). Demonstrated in zinc-air batteries with 400 h durability at 10 mA·cm⁻².

4. Qiu, Y.; **Zhang, Y.**; Xiao, S.; Tian, J.; Zhang, R.; Sun, L.; Tan, K. B.; Duh, Y.-S.; Li, Q.; Huang, J.; Fu, G.* Low-coordinate Cu₁-N₂ sites in single-atom catalyst for highly efficient propylene epoxidation with molecular oxygen. *Nat. Catal.*, under revision.

Overview: Developed a novel Cu₁-N₂ single-atom catalyst for direct propylene epoxidation, achieving record PO formation rates. Advanced characterization and DFT studies revealed exclusive Cu-N coordination enhances electrophilic oxygen species formation, enabling selective C=C bond activation.

5. Liu, X.; **Zhang, Y.**; Li, Y.; Yan, B.; Zhao, X.* Facile preparation of 2D nanomaterials by chelating agent-assisted electrodeposition for efficient electrochemical oxidation of biomass-derived aldehydes coupled with CO₂ reduction. *Appl. Catal. B: Environ.*, **2025**, 368, 125146.
6. Li, Y.; Ouyang, D.; Liu, X.; **Zhang, Y.**; Niu, Z.; Zhu, J. Y.; Pan, X.; Zhao, X.* Self-powered electrochemical synthesis of hydrogen peroxide from air and lignin. *Energy Environ. Sci.*, **2025**, 18, 3633-3646.
7. Dai, Z.; Liu, X.; Liu, N.; **Zhang, Y.**; Zhao, X.* Upgrading biomass derived furan aldehydes by coupled electrochemical conversion over silver-based electrocatalysts. *Chem. Eng. J.*, **2024**, 488, 151001.
8. Xiao, S.; Wei, C.; Li, R.; Qiu, Y.; Tan, K. B.; **Zhang, Y.**; Huang, J.*; Li, Q. Supported AgCuCl Nanoclusters as Bimetal Catalysts for Propylene Epoxidation with Molecular Oxygen. *ACS Appl. Nano Mater.*, **2023**, 6(11), 9687-9696.

SUMMARY

- Top ranking in Chemical Engineering at Xiamen University and Tsinghua University, with excellent grades
- Familiar with a variety of materials synthesis and characterisation tools, such as XAS, XRD, SEM, TEM, etc.
- Proficient in MS, VASP, CP2K, Gaussian, VESTA, vaspkit, and other quantum chemistry calculation software and scripts
- Experienced in both experimental and theoretical work, skilled in communicating with experimental collaborators
- Proficient in MATLAB, Python, C, and other programming languages
- A team player, able to work under pressure and have a high level of execution